SIMATS ENGINEERING ASSESSMENT TOOL 2: Concept Mapping

COURSE NAME: Cloud Computing and
Big Data Analytics
COURSE CODE: CSA15
COURSE OUTCOME COVERED CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)
Assessment Tool Weightage: 20%
Dave's Taxonomy: Imitation (Level 1) Question Count: 4
Total Marks: 20
Code of Conduct I Tharunkumar.S / 192511416
(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the student: _____
Tharunkumar.S
Assessment Tasks Concept Mapping Tasks Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy. 1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations. 2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution,
distributed systems, and service delivery. 3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user
responsibilities. 4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support
cloud deployment and scalability.
Expected concept nodes: Elasticity, scalability, resource pooling, on-demand access, measured service Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer Infrastructure layer, platform layer, application layer, shared responsibility
Concept Mapping Rubric Criteria Weightage Level 4 -
Exemplary
Level 3 - Proficient
Level 2 - Developing
Level 1 - Beginning Concept Identification
25% All major
concepts are accurately identified
Most concepts are identified correctly
Limited concepts are identified
Essential concepts are missing
Relationship 25% Links between Most links are Some links are Relationships are
correct Mapping concepts are
weak or unclear mostly incorrect accurate and meaningful Hierarchy and Organization
No logical organization
Technical Relevance
20% Excellent

Mostly organized
Basic structure structure with
with minor issues
with weak clear hierarchy
hierarchy 20% Map strongly
Mostly aligned to
Partially aligned
Weak alignment reflects cloud
topic
to topic
to topic course content Visual Clarity 10% Neat, readable,
Generally
Somewhat
Poorly presented and easy to
readable
cluttered interpret

Question: 1

cloud

Computing

Fundamentals

CLOUD COMPUTING

is defined has

A model that
has
Provides
also has

M

M
characteristics

Benefits

limitations

on demand

self **service**

Cost efficiency

High

↑

Scalability

Provider on-demand access to shared sy

Computing resources (servers, storage, database & networking) over

the

internet with pay an - me model

- you

Broad **network** Access

Resource Pooling

Rapid elasticity

→ Anywhere

Access

→Internet dependency)

Security & Privacy)

concerns

Vender lock in

Measured

Service

Reliability & availability Fast deploy

-ment

Service outager

Limited

control

infrastructure

COI - AT-2: Concept Mapping

Name i

Coune Code: CSAISI

Reg No: 192511416

Tharunkumas

CONCEPT

MAP 2:

EVOLUTION

OF

cloud

COMPUTING

EVOLUTION

OF

CLOUD COMPUTING

Hardware

Evolution

Mainframe Computer C1960's).

enabled

Internet &

Software Evolution.

is driven by

enabled

Pistributed Betonite systems

yam/2

TCP/IP
Networking
9644049

Personal Computers (1970's -1980's)
World Wide
Web

↓

client-server-systems
(1990's)

Pata centre with
high performance Servers (2000's)

Virtualization and
modern hardware C2010 onward),
Web Applications)

Web Services
& Api's

cloud native application and
SAAS
cluster
computing
led to

Grid Computing

Pistributed
systems (storage))

Parallel Processing
Service
Challery

↓

(utility computing)

↓

Infrastructure As Service (I *AAS*)

I "Platform as

Service (Paas),

Softwand cul
Service (saas)

Reloure sharing
Anything *as* a
Service (xaas)
across Multiple
noder

results in

Modern cloud E Computing

CONCEPT

MAP

लहर

CLOUD SERVICE

MODELS

CLOUD
SERVICE MODELS

Taas (Infrastructure, As Service)

Paas (Platform

As fervices

Provide

manager

Saas (Software

As **Service)**

Provide

Manager

Xaas (**Anything** as Service)

Provide

manager

Infrastructure

Infrastructure,

(**server**, storage

networking

os, runtime, middleware

Application, Platform, Infrastructure

virtualization)

User Manager

Operating System, Application, data

User

Manager

Application, code, data & Configuration

User

Manager

Data acces **user** settinge, permissions

Provider Manage

Specific service and 3

underlying components

User

Manager

service, usage. configuration & data

CONCEPT
MAP
VIRTUALIZATION & WEB SERVICES
AS
CLOUD
ENABLERS

Virtual
Hypervisor
SOAP
REST
Machine (VM))
(eg: VMware, Hyper-*V)*
(KML Bared)
CHTTP
-based,
runson
Creates & manager

Virtualization
C Resource abstraction)

Revoure
Pooling
Chicker
hardware)
Abstraction
Scalability Crapid

C204706 7007

Provisioning

Provider Cpublisher

senice)
enables

ZFRAICE

Dynamic
Rolource **Provisiony**
High..
Scalability,
HODETZ

cloud computing (Deployment &
Scalability)

Provide
are protocols of

↓

Web Services

(Service Communication

via
api's)

API
Interface

enables

On demand
access

Efficient **refovice** utilization

Consumer Cuses
service